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DIV:SE1(S13)

Pandas Methods:

Methods:

```
import pandas as pd

# 1. Load CSV File
df = pd.read_csv("myfolder/customers.csv")

print("\n===== DATASET LOADED =====")

# 2. Display First 5 Rows
print("\n1. First 5 Rows")
print(df.head())

# 3. Display Last 5 Rows
print("\n2. Last 5 Rows")
print(df.tail())

# 4. Shape of Dataset
print("\n3. Shape of Dataset")
print(df.shape)
```

Output:

```
1. First 5 Rows
      customer_id  became_member_on  gender  age  income
0  68be06ca386d4c31939f3a4f0e3dd783    20170212   NaN  118     NaN
1  0610b486422d4921ae7d2bf64640c50b    20170715     F   55  112000.0
2  38fe809add3b4fcf9315a9694bb96ff5    20180712   NaN  118     NaN
3  78afa995795e4d85b5d9ceeca43f5fef    20170509     F   75  100000.0
4  a03223e636434f42ac4c3df47e8bac43    20170804   NaN  118     NaN

2. Last 5 Rows
      customer_id  became_member_on  gender  age  income
16995  6d5f3a774f3d4714ab0c092238f3a1d7    20180604     F   45  54000.0
16996  2cb4f97358b841b9a9773a7aa05a9d77    20180713     M   61  72000.0
16997  01d26f638c274aa0b965d24cefe3183f    20170126     M   49  73000.0
16998  9dc1421481194dcd9400aec7c9ae6366    20160307     F   83  50000.0
16999  e4052622e5ba45a8b96b59aba68cf068    20170722     F   62  82000.0

3. Shape of Dataset
(17000, 5)

4. Column Names
Index(['customer_id', 'became_member_on', 'gender', 'age', 'income'], dtype='str')
```

Methods:

```
# 5. Column Names
print("\n4. Column Names")
print(df.columns)

# 6. Dataset Information
print("\n5. Dataset Info")
print(df.info())

# 7. Check Missing Values
print("\n6. Missing Values")
print(df.isnull().sum())

# 8. Remove Missing Values
df_clean = df.dropna()

print("\n7. Dataset After Removing Null Values")
print(df_clean.head())

# 9. Fill Missing Values
df_fill = df.fillna(0)

print("\n8. Fill Missing Values with 0")
print(df_fill.head())
```

Output:

```
5. Dataset Info
<class 'pandas.DataFrame'>
RangeIndex: 17000 entries, 0 to 16999
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype  
---  --
0   customer_id           17000 non-null  str    
1   became_member_on      17000 non-null  int64  
2   gender                14825 non-null  str    
3   age                  17000 non-null  int64  
4   income                14825 non-null  float64
dtypes: float64(1), int64(2), str(2)
memory usage: 664.2 KB
None

6. Missing Values
customer_id           0
became_member_on      0
gender                2175
age                   0
income                2175
dtype: int64
```

```

7. Dataset After Removing Null Values
      customer_id  became_member_on  gender  age  income
1  0610b486422d4921ae7d2bf64640c50b  20170715  F  55  112000.0
3  78afa995795e4d85b5d9ceeca43f5fef  20170509  F  75  100000.0
5  e212756f4f64592b11af22de27a7932  20180426  M  68  70000.0
8  389bc3fa690240e798340f5a15918d5c  20180209  M  65  53000.0
12 2eeac8d8feae4a8cad5a6af0499a211d  20171111  M  58  51000.0

```

```

8. Fill Missing Values with 0
      customer_id  became_member_on  gender  age  income
0  68be06ca386d4c31939f3a4f0e3dd783  20170212  0  118  0.0
1  0610b486422d4921ae7d2bf64640c50b  20170715  F  55  112000.0
2  38fe809add3b4fcf9315a9694bb96ff5  20180712  0  118  0.0
3  78afa995795e4d85b5d9ceeca43f5fef  20170509  F  75  100000.0
4  a03223e636434f42ac4c3df47e8bac43  20170804  0  118  0.0

```

```

9. Statistical Summary
      became_member_on      age      income
count      1.700000e+04  17000.000000  14825.000000
mean        2.016703e+07    62.531412   65404.991568
std          1.167750e+04    26.738580   21598.299410
min          2.013073e+07    18.000000   30000.000000
25%          2.016053e+07    45.000000   49000.000000
50%          2.017080e+07    58.000000   64000.000000
75%          2.017123e+07    73.000000   80000.000000
max          2.018073e+07   118.000000  120000.000000

```

Methods:

```

# 10. Statistical Summary
print("\n9. Statistical Summary")
print(df.describe())

# 11. Select Specific Columns
print("\n10. Select Gender and Income Columns")
print(df[['gender', 'income']].head())

# 12. Filter Data
print("\n11. Filter Age Greater Than 50")
print(df[df['age'] > 50].head())

```

Output:

```

10. Select Gender and Income Columns
      gender  income
0      NaN      NaN
1      F  112000.0
2      NaN      NaN
3      F  100000.0
4      NaN      NaN

11. Filter Age Greater Than 50
      customer_id  became_member_on  gender  age  income
0  68be06ca386d4c31939f3a4f0e3dd783  20170212  NaN  118      NaN
1  0610b486422d4921ae7d2bf64640c50b  20170715  F  55  112000.0
2  38fe809add3b4fcf9315a9694bb96ff5  20180712  NaN  118      NaN
3  78afa995795e4d85b5d9ceeca43f5fef  20170509  F  75  100000.0
4  a03223e636434f42ac4c3df47e8bac43  20170804  NaN  118      NaN

12. Sort Data by Income
      customer_id  became_member_on  gender  age  income
2924  b624317c94fc496288b746f9be43d170  20160213  F  84  120000.0
11502  0a947767586e4587b06b8ca3efc3c8e7  20180512  M  90  120000.0
2984  731b3f8dd6e2404cb9d7dd508e4c43f1  20171119  F  88  120000.0
15450  5a56cfe9903e420d965a451e84f2bf69  20160403  M  66  120000.0
3843  1a8e25a2abe04bc4aa13f91b45c369db  20160825  M  81  120000.0

```

Methods:

```

# 13. Sorting Data
print("\n12. Sort Data by Income")
print(df.sort_values(by='income', ascending=False).head())

# 14. Group By Operation
print("\n13. Average Income by Gender")
print(df.groupby('gender')['income'].mean())

# 15. Add New Column
df['income_tax'] = df['income'] * 0.10

print("\n14. New Column Added (Income Tax)")
print(df.head())

# 16. Unique Values
print("\n15. Unique Gender Values")
print(df['gender'].unique())

```

Output:

```

13. Average Income by Gender
gender
F    71306.412139
M    61194.601603
O    63287.735849
Name: income, dtype: float64

14. New Column Added (Income Tax)
   customer_id  became_member_on  gender  age  income  income_tax
0  68be06ca386d4c31939f3a4f0e3dd783    20170212    NaN   118      NaN         NaN
1  0610b486422d4921ae7d2bf64640c50b    20170715     F    55  112000.0    11200.0
2  38fe809add3b4fcf9315a9694bb96ff5    20180712    NaN   118      NaN         NaN
3  78afa995795e4d85b5d9ceeca43f5fef    20170509     F    75  100000.0   10000.0
4  a03223e636434f42ac4c3df47e8bac43    20170804    NaN   118      NaN         NaN

15. Unique Gender Values
<StringArray>
[nan, 'F', 'M', 'O']
Length: 4, dtype: str

```